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Ionic wave propagation and collision in an excitable circuit model of microtubules

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In this paper, we report the propensity to excitability of the internal structure of cellular microtubules, modelled as a relatively large one-dimensional spatial array of electrical units with nonlinear resistive features. We propose a model mimicking the dynamics of a large set of such intracellular dynamical entities as an excitable medium. We show that the behavior of such lattices can be described by a complex Ginzburg-Landau equation, which admits several wave solutions, including the plane waves paradigm. A stability analysis of the plane waves solutions of our dynamical system is conducted both analytically and numerically. It is observed that perturbed plane waves will always evolve toward promoting the generation of localized periodic waves trains. These modes include both stationary and travelling spatial excitations. They encompass, on one hand, localized structures such as solitary waves embracing bright solitons, dark solitons, and bisoliton impulses with head-on collisions phenomena, and on the other hand, the appearance of both spatially homogeneous and spatially inhomogeneous stationary patterns. This ability exhibited by our array of proteinic elements to display several states of excitability exposes their stunning biological and physical complexity and is of high relevance in the description of the developmental and informative processes occurring on the subcellular scale. *Published by AIP Publishing.*

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The existence of a mechanical signal due to some applied centrifugal force, though marginal, has been suspected to be responsible of the positional information in cortical microtubules. On the same footing, their electrical activities due to a constant state of flux, continually polymerize and depolymerize them (as observed in animal cells), unless stabilized by the binding of guanosine triphosphate to the ends, which inhibits depolymerization. The dynamics and nature of the information processes underlying these functional mechanisms at the subcellular scale are still not completely understood. We have developed an electrical model with nonlinear resistive properties that can model the excitability of these biological systems. This tunable excitable model, where the dispersion, nonlinearity, and dissipation can be easily regulated, demonstrates rich dynamics, and can serve as a tool for studying the physiological complexity of cortical microtubules.

oxidation of CO on platinum catalysts.³ They equally incorporate biological paradigms such as intracellular waves of calcium induced calcium release, head-on collisions of waves in excitable FitzHugh-Nagumo systems,⁴ the electrical membrane excitation in nerve and muscular tissues (relevant in cardiac arrhythmia for instance), and some simple morphogenetic systems,⁵ to name just a few.

Nowadays, biological systems have been the core components in the development of nonlinear science due to their outstanding complexity both at small and large scales. Their baroque complexity then requires some strong computational treatments safely guided by mathematical insights, based on simple nonlinear systems, or rationally reduced models. Starting from the ground that biological systems usually emerge as a large collective set of topologically connected (all-to-all, sparsely, or nearest neighbors interactions) biochemical units (cells), their computational modelling, crucial for their quantitative and qualitative dynamical studies, forms a key ingredient of the interpretation of the mathematical and experimental outcomes obtained from their inspections. Structurally speaking, the basic building block of the architecture of any biological system is the cell. It is mainly made up of a substance called the cytoplasm, containing 80% of water in its composition. Throughout the cytoplasm of eukaryotic cells (cells that have membrane-bound organelles such as nucleus and endoplasmic reticulum) the cytoskeleton, which comprises tubular polymers called microtubules, is found.^{6,7} Microtubules play some key roles in living organisms. They are triggered by the signalling processes occurring within the cellular milieu. Microtubules

I. INTRODUCTION

During the last decades, quite a sizable amount of mathematical treatment has been performed in the field of complex nonlinear systems as a whole and in the area of nonlinear excitable media, in particular. Typical examples of such systems include chemical excitable milieu, namely, the Belousov-Zhabotinsky reaction^{1,2} and the reaction of

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play some crucial roles in a number of cellular processes and are involved in maintaining the structure of the cell. Together with microfilaments and intermediate filaments, they form the cytoskeleton. They also make up the internal structure of cilia and flagella which play a great role in movement.^{7–10} The act of withdrawing the hand when it comes into contact with a hot object, the act of an organism displacing itself to locate food when in need, and many other occurrences are evidence of some degree of information processing and transmission within the internal structure of living beings. These processes usually occur in the form of wave phenomena.² Such information or impulse called ionic wave is in the form of an electric field. This ionic wave follows a pattern that sometimes emerge in the form of a localized wave, solution of a widespread class of weakly non-linear dispersive partial differential equations referred to as non-linear localized waves of soliton types.^{11,12}

References 13–15 model microtubule as a nanobioelectronic nonlinear circuit possessing the polyelectrolyte features of cylindrical biopolymers, with the assumption that microtubules play a fundamental role in sub-cellular computation and signalling via electronic and protonic conductivity and can then be made useful in hybrid material that offer novel electronic characteristics. Based on these considerations, the nature of the wave solution (or energy) resulting from the model is observed to be a solitonic pulse. Recently, on the same footing, the authors in Ref. 16 investigated the phenomena of wave transmission and amplification in a similar electrical model of microtubules. Therein, a comparison is made between the results obtained on one hand in the absence of a negative nonlinear resistance, and on the other hand, with those obtained in its presence. The signal was found to be decreasing in the former case and to be amplified in the latter.

In this work, considering the strong intrinsic nonlinear character of these biological systems, we consider an electrical model of microtubule with a nonlinear resistance at the level of each electrical unit. This is motivated by the fact that the previous models failed to display some features of excitability as it has been experimentally reported to exist to some extent in microtubules.^{17–26} This consideration accounts for the fact that the processes of opposition to the flow of calcium ions within and between microtubules are more than just linear. We believe that the fluctuations in the internal resistance of the dynamical unit, occurring in the course of the information processing mechanisms occurring along the structure of the microtubule, and which depend to a great extent on the content of the microtubule, may impact significantly on the nature of the physical and biochemical solicitations of these biological systems. Therefore, the aim of this work is to investigate under the aforementioned circumstances (conditions), the related liability of these biological systems to display some interesting features of excitable media. The main goal of this exploration is to throw more light on the puzzling processes pertaining in the robust internal functional activities of cells, which have relevance in maintaining homeostasis in living beings, via stable internal operating conditions, crucial for the perpetuation of life.

This article is organized as follows. In Sec. II, we present our electrical model of microtubule (MT) with nonlinear resistive elements. This model aims at emulating the dynamics of the voltage stimulus, compelling calcium ions to move within and in between microtubule protofilaments (PF). Subsequently, we derive a discrete differential equation of motion for the voltage stimulus of the calcium ions, following some circuit analysis principles. At this stage, we propose ourselves to solve this equation while seeking for solitary waves solutions. For this purpose, we shall use a perturbation technique in the form of the multiple scales expansion method with semi-discrete approximation. The description of this method is coarsely provided, and its development related to the issue at hand is as well expounded. Afterward, an equation of evolution for the ionic wave amplitude is obtained in the form of a complex Ginzburg-Landau equation (CGLE), and the Benjamin-Feir stability analysis of the plane waves solutions of this equation is conducted numerically. In Sec. III, we analytically examine the stability of the plane waves solutions of the CGLE based on the Eckhaus stability analysis approach. The outcomes of this analytical inquiry are compared to those obtained numerically in Sec. II in order to check for their consistency. In Sec. IV, based on the Nozaki-Bekki approach, we propose an analytical solution for the solitary wave solution of our CGLE, whose characteristics will be analyzed in the entire computational domain. Next, we perform the analytical resolution of our CGLE based on the harmonic balance method, in order to work toward the determination of exact travelling symmetric-like ionic waves, solutions of our nonlinear equation. Section V is dedicated to the direct numerical treatment of the main discrete voltage equation of motion, while assuming Ansatz as the exact analytical solution obtained in Sec. IV. Finally, Sec. VI concludes the work.

II. THE ELECTRICAL CIRCUIT MODEL OF MICROTUBULES AND THE WAVE AMPLITUDE EQUATION

A. The model

Microtubules (MTs) are essential components of the cytoskeleton, found throughout the cytoplasm of eukaryotic cells. They are hollow cylinders made up of protofilaments (PF) representing series of proteins known as tubulin dimers. Each PF is assimilated to a succession of N adjacent electrical units. Each unit cell number n of the lattice is constituted of a linear inductance L in series with a linear resistance r , shunted by a non-linear capacitance $C(V_n)$ in parallel with a non-linear resistance R_n .¹⁶ In standard models, the non-linear components are only represented by a conductance and a capacitance.^{9,14,27,28} Most mathematical models for transmitters^{27,29} add a cubic non-linear resistance for ionic current flow through nano-pores of neurons. Freedman *et al.*²⁷ observed that the current-voltage characteristic of the MT lumen resistance in the presence of cations can be fitted by a cubic polynomial. Thus, a non-linear resistance was suggested for modelling cation flow through nano-pores of MT.¹⁶ Priel *et al.*,^{16,30} in an experiment, isolated

microtubules and infused electrical signals through one end. Overwhelmed by the reception of the emitted signal at the other end of the microtubules, they could ascertain the fact that microtubules can be looked upon and described as biological conducting wires, suitable for electrical pulses transmission. Therefore, microtubules play an important role in electrical activities within cells.

It is noteworthy that excitable systems usually appear as spatially extended systems, which as a result of the application of a super-threshold stimulus, are capable of generating and conducting nonlinear waves of excitations. These regenerative waves travel without decrement as a solitary propagating wave or wave train. In this case, subsequent to the excitation, the medium needs some time to recover its excitability. During this time interval called the refractory period, the medium cannot be excited again. Thus, based on these facts, our proposed electrical model of microtubule should be consistent with these features of excitable media. In an attempt to do so, we suggest below in Fig. 1 an electrical circuit model for the microtubules, with a variety of specifications, including nonlinear resistive elements. In this model, we focus mainly on the voltage stimulus V_n urging cations (calcium ions or currents) within and between microtubule protofilaments.

The mathematical description of the model at hand is based on Kirchoff's network rules. The microtubule, being assimilated to a succession of N adjacent unit cells, depicts a transmission line. This transmission line then portrays one protofilament of a microtubule. Each unit cell number, n , of the lattice is made up of a linear inductance, L , in series with a linear resistance r , shunted by a nonlinear capacitance $C(V_n)$ in parallel with a nonlinear resistance R_n . The proposed circuit model combines two models where on one hand the $C(V_n)$ characteristics is a nonlinear function of the voltage,¹⁵ and on the other hand, the resistance is a nonlinear function of the voltage.¹⁶ The characterizations of the origins of the current, capacity, and resistance are well described in Ref. 15.

However, these models were better suited to explore the ability of microtubules to act as transmission lines supporting wave amplification phenomena. They could not display properties of excitability which have been experimentally shown to exist in microtubules.^{17–26} The role played by microtubules in neuronal excitability had been proposed several years ago by Matsumoto and Sakai, based on experiment using squid giant axons which resulted in the direct correlation between excitability and microtubule assembly.^{17–19} In fact, the excitability of the axon membrane, once suppressed by intracellular perfusion with solutions unfavourable for microtubule assembly, can be restored by perfusing the axon

with a solution containing tubulin and its polymerizing reagents.^{19,25} In Ref. 25, Terakawa *et al.*, following experimental observations, suggested that the submembranous cytoskeletons, including microtubules, provide an important mechanical support to the excitable membrane but are not essential elements in channel activities. Also, Ichikawa *et al.* artificially controlled the physiological conditions of axonal microtubules in squid giant axons under an optical microscope by perfusing them intracellularly with artificial solutions and by clamping the membrane voltage to measure membrane currents.²⁶ Their observations enabled them to conclude that axonal microtubules determine the generation and maintenance of membrane excitability in squid giant axons. In the same line, Braak *et al.*²⁰ reported from experimental studies that the microtubule-associated protein Tau is implicated in neurodegenerative disorders and described briefly how the Tau protein may modulate neuronal excitability by interacting with the microtubules at the axon initial segment. The authors in Ref. 9 have hypothesized that microtubules may exhibit some features of excitable media as they support the creation and propagation of some sort of solitonic ionic waves along their axis. In the context of their analysis, the origin of the nonlinearity in microtubules stems from the nonlinear capacitance of each tubulin dimer. Similarly, in this work, it is keynote that the polynomial form of the nonlinearity we use here, as shown in Eq. (2), stems from its electrical charge and is well known to be appropriate in excitable neural models,³¹ in replacement of the previously used logarithmic form which has been shown to have some limitation in mimicking enough the experimentally observed physiological features of excitability of these cellular systems.

Applying the current rule to the circuit at points A and B of the circuit, and the voltage rule to the loops BCRQB and ABQPA, where the entire transmission line represents a protofilament of a microtubule and V_n denotes the voltage stimulus of the calcium ions (cations), we obtain the equation

$$V_{n+1} - 2V_n + V_{n-1} = L \frac{d^2 Q_n}{dt^2} + 3B_1 L V_n^2 \frac{dV_n}{dt} - B_2 L \frac{dV_n}{dt} + r \frac{dQ_n}{dt} + rB_1 V_n^3 - rB_2 V_n, \quad (1)$$

where Q_n denotes the quantity of charge that flows through the n_{th} unit cell (i.e., the number of calcium ions driven across the n_{th} unit cell multiplied by the quantity of charge in one calcium ion). $C(V_n)$ denotes the nonlinear capacitance element, which represents quantity of charge urged across the n_{th} unit cell per unit voltage. $C(V_n)$ possesses an electrical charge, Q_n , which takes the form

$$Q_n = C_0(1 - \alpha V_n - \beta V_n^2)V_n, \quad (2)$$

obtained by using the form of the charge proposed in Ref. 31.

Here, C_0 is the capacitance in the linear regime and α and β are positive parameters that account for non-linearity in the charge.

The nonlinear resistance effects are characterized by the relation

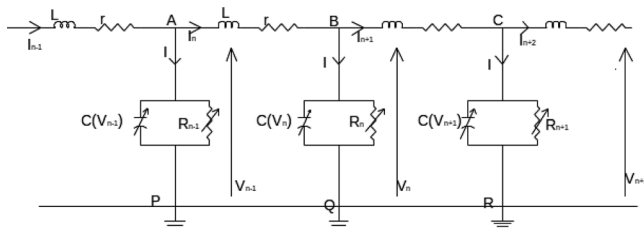


FIG. 1. The electrical circuit model of microtubule.

$$I_{R_n} = f(V_n) = B_1 V_n^3 - B_2 V_n, \quad (3)$$

that is, I_{R_n} is the current flowing through the non-linear resistance R_n ; B_2 is a positive parameter that accounts for linearity in the current; and B_1 is a positive parameter that accounts for non-linearity in the current.¹⁶ Equation (3) is approximately the I-V relationship of the nonlinear resistance and the nonlinear differential equation [Eq. (1)] becomes

$$\ddot{V}_n - \Omega^2 V_n + (\gamma_0 + \gamma_1 V_n + \gamma_2 V_n^2) \dot{V}_n + (\eta_1 + \delta V_n) \dot{V}_n^2 + (\eta_1 V_n + \lambda V_n^2) \ddot{V}_n + \eta_2 V_n^3 = D_0 (V_{n+1} - 2V_n + V_{n-1}), \quad (4)$$

where

$$\Omega^2 = \frac{rB_2}{LC_0}, \quad \gamma_0 = \frac{r}{L} - \frac{B_2}{C_0}, \quad \gamma_1 = -2\frac{\alpha r}{L}, \quad \gamma_2 = 3\left(\frac{B_1}{C_0} - \frac{\beta r}{L}\right), \\ \eta_1 = -2\alpha, \quad \delta = -6\beta, \quad \lambda = -3\beta, \quad \eta_2 = \frac{rB_1}{LC_0}, \quad D_0 = \frac{1}{LC_0}.$$

Equation (4) describes the spatiotemporal evolution of the voltage $V_n(t)$. The terms containing B_1 and B_2 are the contribution of the ionic flow. The terms with r represent the contribution of the linear resistance r . After the equality sign, we have the coupling entity between nearest neighbors $(n-1)$, n , $(n+1)$ of the dimer lattice. The system is dimensionless and it is needless to insist on space and time units. An approximate solution to Eq. (4) can be obtained using perturbation methods. In order to do so, we introduce the variables

$$V_n = \epsilon \phi_n,$$

where $\epsilon \ll 1$.

We consider that γ_0 is a perturbed parameters of order ϵ^2 . This perturbation arises from the fact that we are looking for solutions in a weakly dissipative medium. Keeping the first two nonlinear terms of the development, the equation becomes

$$\ddot{\phi}_n - \Omega^2 \phi_n + \epsilon(\epsilon\gamma_0 + \gamma_1 \phi_n + \epsilon\gamma_2 \phi_n^2) \dot{\phi}_n + (\epsilon\eta_1 + \epsilon^2 \delta \phi_n) \dot{\phi}_n^2 + (\epsilon\eta_1 \phi_n + \epsilon^2 \lambda \phi_n^2) \ddot{\phi}_n + \epsilon^2 \eta_2 \phi_n^3 = D_0 (\phi_{n+1} - 2\phi_n + \phi_{n-1}). \quad (5)$$

The resolution of Eq. (4) reveals the nature of the voltage signal within and between dimer molecules. However, its solution is obtained through the use of Eq. (5).

It is keynote to mention that several authors investigated the mechanical and electrical features of microtubules.^{6-10,13-28,30,32-35} Indeed, due to their electrical properties, microtubules are known to play important roles in many cellular functions including motility, mitosis, transport, and the maintenance of cell shape by forming cytoskeletons.^{30,32,33} Nerve tissues contain a large amount of tubulin and microtubules, and experimental evidence has been accumulated in support of the view that neuronal microtubules are directly involved in axonal transport.^{34,35} Motivated by these facts, Sekulic *et al.*³² suggested that microtubules (by

for instance the collective dynamics of the flexible C-terminal tails decorating its outer surface) could crucially act as a mechanical support of localized waves aimed for communication between microtubules and associated proteins.

B. The wave amplitude equation

At this stage, let us reiterate the fact that our major concern is to investigate for the existence of low-amplitudes non-linear ionic waves in a microtubule. For this purpose, it is convenient to use the multiple scale expansion method which can be applied with a semi-discrete approximation in order to find the solution to our differential equation. The semi-discrete approximation is a perturbation technique in which the carrier waves are kept discrete while the amplitude is treated in the continuum limit. Applying this method allows one to study the modulation of a plane wave caused by nonlinear effects. We proceed by making a change of variables according to the new time and space scales $U_n = \epsilon^n u$ and $T_n = \epsilon^n t$. It should be noted that the principal purpose of using the multiple scale expansion is to find a solution $V(u, t)$ depending on these new sets of variables as a perturbation series of functions. We will consider here that

$$V(u, t) = \sum_{n=1}^{\infty} \epsilon^n \phi_n(U_0, U_1, U_2, \dots, T_0, T_1, T_2, \dots), \quad (6)$$

where each U_n or T_n is treated as an independent variable. Thus, we also obtain a perturbation series of operators from both independent variables. The temporal derivatives can be expressed as

$$\frac{\partial}{\partial t} = \frac{\partial T_0}{\partial t} \frac{\partial}{\partial T_0} + \frac{\partial T_1}{\partial t} \frac{\partial}{\partial T_1} + \frac{\partial T_2}{\partial t} \frac{\partial}{\partial T_2} + \dots \quad (7)$$

Similarly, the spatial derivatives can be expressed as

$$\frac{\partial}{\partial u} = \frac{\partial U_0}{\partial u} \frac{\partial}{\partial U_0} + \frac{\partial U_1}{\partial u} \frac{\partial}{\partial U_1} + \frac{\partial U_2}{\partial u} \frac{\partial}{\partial U_2} + \dots \quad (8)$$

An important feature of this method is that the solution of the original problem will only be obtained if the multidimensional space generated by the new set of variables U_n and T_n comes from the physical line.¹¹ Worth noting is the fact that these auxiliary variables permit us to impose appropriate conditions to the system thus assuring that the asymptotic expansion converges uniformly for small values of ϵ .³⁹

In order to solve the equation of motion of the voltage stimulus, we consider the following form of solution (ansatz) for the dynamics of our discrete electrical circuit model of microtubule:

$$\phi_n = F_n e^{j\theta_n} + F_n^* e^{-j\theta_n} + \epsilon(G_n + H_n e^{2j\theta_n} + H_n^* e^{-2j\theta_n}) + O(\epsilon^2), \quad (9)$$

with $\theta_n = qn - wt$, where q is the normal mode wave vector, n denotes the unit cell number in the dimer lattice, and w is the angular velocity of the wave.

The fact that there are nonlinear terms in Eq. (5) leads one into the prediction that through frequency superpositions, the first harmonics of the wave will contain terms in

$e^{\pm 2j\theta_n}$ as well as terms without any exponential dependence. We will equally consider that the amplitudes F , G , and H change slowly in space and time. This is formalized in a multiple scale expansion in a method referred to as semi-discrete approximation because it preserves a carrier wave which obeys a discrete dispersion relation and only the variation of the amplitude is treated in the continuum limit.

In this semi-discrete approximation, F , G , and H are set to be independent of the “fast” variables t and u , but to depend on the “slow” variables $U_1 = \epsilon u$, $U_2 = \epsilon^2 u$, $T_1 = \epsilon t$, and $T_2 = \epsilon^2 t$. Applying now the continuum limit approximation on these amplitudes will yield $F_n(t)$, $G_n(t)$, and $H_n(t)$ becoming $F(U_1, U_2, T_1, T_2)$, $G(U_1, U_2, T_1, T_2)$, and $H(U_1, U_2, T_1, T_2)$, respectively. $F_{n\pm 1}$ is obtained up to ϵ^2 by a Taylor expansion

$$F_{n\pm 1} = F \pm \epsilon \frac{\partial F}{\partial U_1} \pm \epsilon^2 \frac{\partial F}{\partial U_2} + \frac{\epsilon^2}{2} \frac{\partial^2 F}{\partial U_1^2} + O(\epsilon^3).$$

Similar equations are written for G_n and H_n . These together with Eq. (9) are inserted into the two equations depicting our model, that is, Eq. (5). We seek relations for which terms in $e^{\pm j\theta_n}$, $e^{\pm 2j\theta_n}$ and terms without exponential dependence cancel out at different orders of ϵ (at different orders of perturbation).

At the zeroth order of perturbation, we obtain the dispersion relation of linear waves of the system described by Eq. (5)

$$w^2 = -\Omega^2 + 4D_0 \sin^2 \frac{q}{2}. \quad (10)$$

By definition, the coefficients, B_1 , B_2 , r , L , and C_0 are positive constants. Therefore, Eq. (10) is valid if the wave number satisfies the condition¹⁶

$$q \geq q_c = 2 \arcsin \sqrt{\frac{\Omega^2}{4D_0}}. \quad (11)$$

It follows from Eq. (11) that the system is a low band pass filter with the allowed band defined by the frequency interval $[0, \omega_c]$, where $\omega_c^2 = -\Omega^2 + 4D_0$ is the square of the upper cut-off frequency.¹⁶ The dispersion relation, Eq. (10), is indeed that of a discrete model and thus meets up with expectations. Figure 2 presents the graph of this dispersion relation. It depicts a scenario whereby there are limited intervals of values of the wave number which are suitable for the existence of linear waves within the medium. These epochs of values of q suitable for existence of linear waves are regularly separated by forbidden intervals of values of the wave number. These forbidden gaps then naturally indicate that linear waves are not accessible for the system for some given values of q .

At the first and second orders of perturbation, we obtain the coefficient

$$G = 0, \quad (12)$$

the relation

$$H = - \left[\frac{\omega D_0 (2\omega \eta_1 + j\gamma_1)}{D_0 (\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)} \right] F^2, \quad (13)$$

the group velocity of the ionic waves conveyed by our model

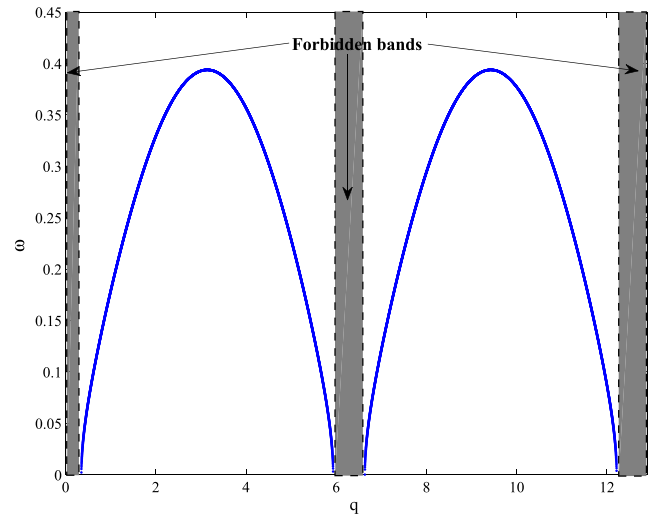


FIG. 2. The dispersion relation of the ionic wave. $D_0 = 0.04$ and $\Omega^2 = 0.0049$.

$$v_g = \frac{D_0 \sin q}{w}, \quad (14)$$

and at higher order, the equation of evolution of the solitonic function

$$j \frac{\partial F}{\partial \tau_2} + \frac{P}{2} \frac{\partial^2 F}{\partial \xi_1^2} + Q |F|^2 F + j \frac{R}{2} F = 0, \quad (15)$$

where we have used the rescaling (change of variables)

$$\xi_n = U_n - v_g T_n, \quad \tau_n = T_n. \quad (16)$$

Equation (15) shows that the evolution of ionic waves in this electrical circuit model of microtubules is described by the Complex Ginzburg-Landau equation (CGLE) where the coefficients of dispersion and dissipation, P and R , respectively, are real, while the coefficient of nonlinearity, Q , is complex. It can be seen that the wave propagates in a frame of reference that travels with velocity, v_g (the group velocity), where

$$P = \frac{D_0 w^2 \cos q - D_0^2 \sin^2 q}{w^3}, \quad (17)$$

$$Q = Q_r + jQ_i, \quad (18)$$

$$R = \gamma_0, \quad (19)$$

where Q_r and Q_i are the real and imaginary parts of the nonlinearity coefficient, respectively, such that

$$Q_r = \frac{\omega D_0}{2} \left(\frac{2\omega^2 \eta_1^2 - 3\gamma_1^2}{D_0 (\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)} - \frac{3\eta_2 + \omega(\delta - 3\lambda)}{2\omega} \right), \quad (20)$$

$$Q_i = \frac{\omega D_0}{2} \left(\frac{7\omega \gamma_1 \eta_1}{D_0 (\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)} \right) + \frac{\gamma_2}{2}. \quad (21)$$

The variation of the coefficient of dispersion P , the real and imaginary parts of the coefficient of nonlinearity Q_r and

Q_i , and the product PQ_r , as functions of the wave number q , are plotted in Fig. 3.

When the dissipation is not considered in some cases, the Complex Ginzburg-Landau equation reduces to the Nonlinear Schrödinger equation.⁴⁰ However, that is not the case in this model because if the dissipation is neglected, the nonlinearity coefficient will still have complex and imaginary parts. Therefore, our amplitude equation will always be described by the Complex Ginzburg-Landau equation.

The dispersion coefficient is real. It follows that the modulational instability depends on the sign of PQ_r . According to Benjamin-Feir instability analysis, plane waves are unstable for positive values of PQ_r whilst they are stable for negative values. Note that this stability criterion does not depend on the manner with which the wave propagates.

In Fig. 3(a), it is observed that the dispersion coefficient P is always negative. Also, in Figs. 3(b) and 3(c), we see that the real and imaginary parts of the coefficient of nonlinearity are always negative and positive, respectively, irrespective of the accessible values of q . In accordance with these observations, as expected from the outcomes of Figs. 3(a) and 3(b), Fig. 3(d) illustrates the fact that the product PQ_r is permanently positive. This suggests that in line with the Benjamin-Feir instability criteria, plane waves' solutions of our biological system are unconditionally unstable. As a result of this, our model of microtubule steadily exhibits a tendency to promote the generation of modulated amplitude waves along the structure of the subcellular lattice. Thus, based on these facts, our biological model depicts an environment purposely conducive for the existence of stable solitary waves, as solutions of our dynamical system. The determination of these soliton solutions in this proteinic network model shall be the core of Secs. III–VI. First of all, in

order to be relatively complete in this survey, Sec. III will be dedicated to analytically ascertain this startling predilection to excitability by the means of the Eckhaus instability analysis of the plane waves solutions of our *CGLE*.

III. NONLINEAR ANALYSIS OF THE ECKHAUS INSTABILITY OF THE MODEL

The complex Ginzburg-Landau equation that describes our model is given in Eq. (15).

A plane wave is a field of energy that propagates with constant frequency and amplitude. An amplitude modulated wave is a wave whose amplitude varies spatiotemporally, while a phase modulated wave is a wave whose phase changes spatiotemporally. In order to study the stability of our wave amplitude equation, we use the following ansatz for our saturated amplitude modulations:^{44,45}

$$F(\xi_1, t) = a(z)e^{i[\phi(z) + \omega_1 t]}, \quad (22)$$

where a and ϕ are real-valued functions of $z = \xi_1 - v_g \tau_2$. Substituting the ansatz in Eq. (22) into Eq. (15), separating the real and imaginary parts and setting $b = a_z$ and $\psi = \phi_z$ yield the set of three coupled nonlinear ordinary differential equations (ODEs)

$$\begin{cases} a_z = b \\ b_z = \frac{2\omega}{P}a + \frac{2}{P}a\psi + a\psi^2 - \frac{2}{P}a^3Q_r \\ \psi_z = \frac{2v_g}{P}\frac{b}{a} - \frac{R}{P} - 2\frac{b\psi}{a} - \frac{2Q_i}{P}a^2. \end{cases} \quad (23)$$

Therefore, the system of Eq. (23) describes the dynamics of amplitude-modulated waves in our electrical circuit model of microtubules.

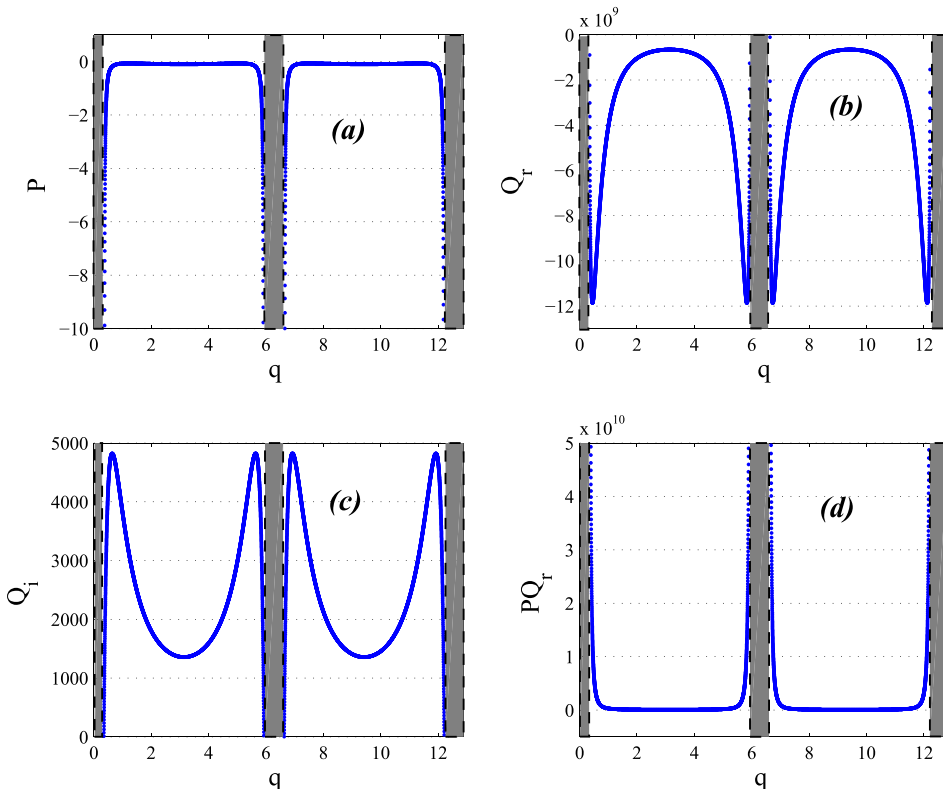


FIG. 3. Variations of the coefficient of dispersion P , the real and imaginary parts of the coefficient of nonlinearity Q_r and Q_i , and the product PQ_r , as functions of the wave number q . The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40\,000$, $\gamma_2 = 5.0$, $\lambda = 0.05$, and $\delta = 0.8$.

We proceed to perform a stability analysis of the fixed points of our model. Since the equilibria of dynamical systems are not always stable, it behooves us to characterize the stability of the fixed points pertaining to our model due to the fact that stable and unstable equilibria play quite different roles in the dynamics of a system. If the fixed points are stable, then the plane waves become stable; if the fixed points are unstable, the plane waves become unstable. The equilibrium points are determined by setting all gradients to zero, that is, $a_z = b_z = \psi_z = 0$, and, taking only positive square roots, we find that the procedure starts from the fixed

point $(a, b, \psi_z) = (\sqrt{\frac{R}{2Q_i}}, 0, \frac{1}{P}[-1 + \sqrt{1 + P(\frac{Q_r}{Q_i}R - 2\omega)}])$, which corresponds to a plane wave solution. In a bid to characterize the stability of the fixed point, we obtain the characteristic polynomial

$$a_0\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0, \quad (24)$$

where

$$\begin{aligned} a_0 &= 1, \quad a_1 = 0, \\ a_2 &= 4\psi^2 + \frac{4\psi}{P} + \frac{6Q_r}{P} - \frac{4v_g\psi}{P} - \frac{4v_g}{P^2} - \psi - \frac{2\psi}{P} - \frac{2\omega}{P}, \\ a_3 &= \frac{4av_g\psi}{P} + \frac{4av_g}{P^2} - 4a\psi^2 - \frac{4a\psi}{P}. \end{aligned} \quad (25)$$

For the fixed point to be stable, all the eigenvalues, λ_k for $k = 1, 2$, and 3 of Eq. (24) must have negative real parts. Using the *Routh-Hurwitz stability criteria*, the necessary and sufficient conditions for stability to occur are

$$a_1 > 0, \quad a_3 > 0, \quad \text{and} \quad a_1a_2 > a_3. \quad (26)$$

The simultaneous fulfillment of these criteria implies that all the eigenvalues of the characteristic equation have negative real parts, synonym of convergence of our dynamical system, following some perturbation deeds, toward the plane waves solutions. Yet, from Eq. (26), $a_1 = 0$. Then, this implies that the Routh-Hurwitz stability criteria can never be concomitantly fulfilled. Thus, the plane waves solutions of our *CGLE* are always unstable. This fact is in perfect agreement with the results obtained numerically in Fig. 3(d) in Sec. II, using the Benjamin-Feir stability condition. Therefore, our *CGLE* admits modulated amplitude waves as only stable solutions. It then suitably describes the dynamics of our lattice of biochemical units as an excitable medium.

IV. NONLINEAR IONIC WAVE SOLITON SOLUTION

A. The Nozaki-Bekki soliton solution

A form of the soliton solution of the amplitude Eq. (15) can be obtained using the Nozaki and Bekki ansatz^{36,38,40,41}

$$F(\xi_n, t) = \frac{F_0 e^\varphi}{1 + e^{(\varphi + \varphi^*)(1+jx)}}, \quad (27)$$

where

$$\varphi = q\xi_n - w\tau, \quad (28)$$

$$\alpha = \beta \pm (2 + \beta^2)^{\frac{1}{2}}, \quad (29)$$

$$\beta = \frac{3Q_r}{2Q_i}. \quad (30)$$

Now, rewriting the ansatz in Eq. (27) by considering the real and imaginary parts and transforming its coefficients following the process in Ref. 37, we arrive at the solution for the calcium wave propagation in the microtubule described by

$$\begin{aligned} V_n &= \epsilon F_0 \left[\frac{e^\varphi \cos(\theta - 2\alpha\varphi) + e^{-\varphi} \cos \theta}{(\cosh 2\varphi + \cos 2\alpha\varphi)} \right] + \epsilon^2 F_0^2 \left[(x \cos 2\theta \right. \\ &\quad \left. - y \sin 2\theta) \left(\frac{e^{2\varphi} \cos 4\alpha\varphi + 2 \cos 2\alpha\varphi + e^{-2\varphi}}{2(\cosh 2\varphi + \cos 2\alpha\varphi)^2} \right) \right] \\ &\quad \left. - \epsilon F_0^2 \left[(x \sin 2\theta - y \cos 2\theta) \left(\frac{e^{2\varphi} \sin 4\alpha\varphi - \sin 2\alpha\varphi}{(\cosh 2\varphi + \cos 2\alpha\varphi)^2} \right) \right] \right], \end{aligned} \quad (31)$$

where

$$x = -\frac{2\omega^2 \eta_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)}, \quad (32)$$

$$y = -\frac{\omega \gamma_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)}. \quad (33)$$

Figure 4 portrays the depiction and behavior this ionic wave in all the computational domains. From Figs. 4(a)–4(c), It is clearly noticed that this analytical solution emerges in the form of a single asymmetric pulse, propagating along the medium. These figures show the three-dimensional spatiotemporal evolution of the travelling ionic impulse [Fig. 4(a)], its upper view [Fig. 4(b)], and its profiles at different times [Fig. 4(c)], respectively. The snapshots of the lattice at several distinct times [as seen in Fig. 4(c)] reveal the fact that the pulse is structurally unstable, as it changes its shape in the course of its propagation through the lattice. Indeed, this asymmetric solution is indicative of the presence of a bright envelope soliton propagating within the one-dimensional spatial array of biochemical units, and whose amplitude is correspondingly depicted in Fig. 4(d) (three-dimensional spatiotemporal evolution), Fig. 4(e) (upper view), and Fig. 4(f) (lattice snapshots at different times).

Equally, it is important to note that the parameter ϵ helps us to differentiate the orders of development of the solitonic pulse. Its effect on the ionic wave can be seen in Fig. 5. This figure represents the snapshots of the lattice structure at the time $t = 46$, for different values of the perturbation, namely, $\epsilon = 0.0001$ [Fig. 5(a)], $\epsilon = 0.001$ [Fig. 5(b)], $\epsilon = 0.01$ [Fig. 5(c)], and $\epsilon = 0.1$ [Fig. 5(d)]. We notice that as the perturbation increases the amplitude of the ionic wave also increases. A similar phenomenon has already been observed in other biological excitable media, namely, in diffusively connected Hindmarsh-Rose neural networks.³⁷ A generalization of this fact is depicted in Fig. 6.

Thus, the previously obtained outcomes show that information processing and transmission along the structure of a

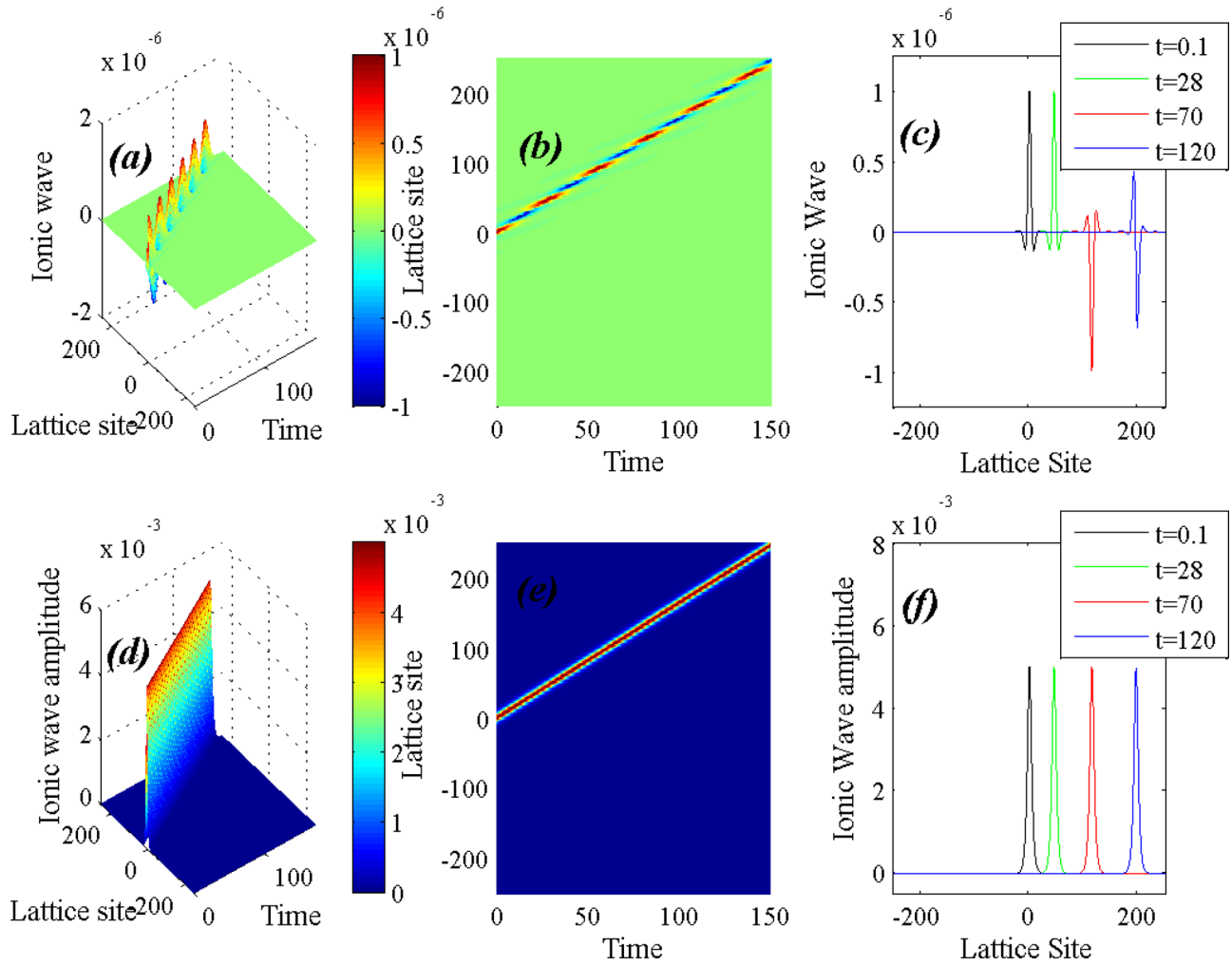


FIG. 4. Spatiotemporal evolution of the ionic wave and its corresponding envelope. The ionic wave is presented in (a) three-dimensional view of the travelling pulse, (b) upper view of the pulse, and (c) ionic wave profiles at different times. The corresponding envelope is depicted in (d) three-dimensional view of the propagating envelope, (e) upper view of the envelope, and (f) envelope profiles at different times. The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40\,000$, $\gamma_2 = 5.0$, $\lambda = 0.05$, $\delta = 0.8$, $q = 1.5$, $F_0 = 0.01$, and $\epsilon = 0.0001$.

microtubule are achievable by means of localized excitations. These solitary waves, because of their robustness and stability, are ideal tools to secure the transmission processes taking place along some relatively large arrays of biological dynamical systems. Their remarkable robustness is predominantly due to their regenerative feature. Yet, one of the challenges usually faced in communication processes is the ability to concomitantly convey not only safely but also quantitatively, a huge diversity of information. This implies the simultaneous transmission of a huge set of harmonics along our biochemical milieu. A typical means to achieve that is to relay the stimulus in the form of a wave packet. Subsequent to these facts, we propose ourselves to seek for wave packets solutions of our nonlinear equation of motion. In an attempt to meet this target, in Sec. IV B, we use the harmonic balance method to solve our wave amplitude equation and derive the expression of the ionic wave propagating in the model.

B. The symmetric solution using the harmonic balance method

Here, we seek exact solutions of the CGLE that pertain to our model. We use the harmonic balance method with a

nonlinear transform to find solutions in terms of Fourier series.⁴² This method, which is a technique used in determining the limit cycle of a dynamical system, will be utilized in this section to obtain the exact travelling solution of the equation describing the dynamics or evolution of our model (the CGLE). In this technique, we define and use a nonlinear transform $l(\xi)$ such that

$$\frac{dl}{d\xi} = f, \quad (34)$$

where $f = f(l) = f(l(\xi))$ is a periodic function of period 2π . Trigonometric functions, which are easier to handle, will be used here.

For our system, we choose

$$f(l) = \cos l. \quad (35)$$

Since the equation describing the evolution of our model, the complex Ginzburg-Landau equation (CGLE), is a complex equation, we consider an ansatz (solution) to it of the form:

$$F(\xi, t) = g(\xi) e^{j[s(\xi) + wt]}, \quad (36)$$

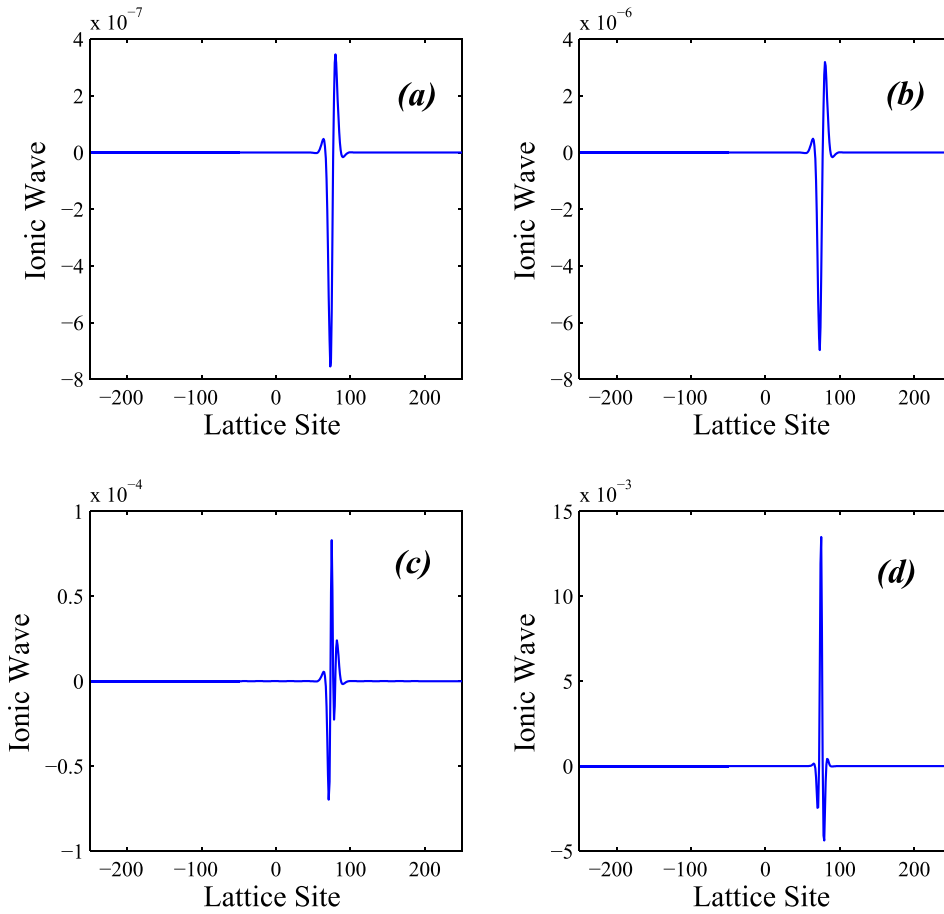


FIG. 5. Effects of small perturbations on the ionic wave. Snapshots of the wave profile taken at $t=46$. (a) $\epsilon=0.0001$, (b) $\epsilon=0.001$, (c) $\epsilon=0.01$, and (d) $\epsilon=0.1$. The parametric values are the same as in Fig. 4.

where $g(\xi)$ and $s(\xi)$ are real functions and $\xi = n_1 u + n_2 t$, n_1 and n_2 being constants.

Inserting Eq. (36) into Eq. (15) yields, for real and imaginary terms

$$g_{\xi\xi} - g s_{\xi}^2 - n_2 n_3 g s_{\xi} + n_4 g^3 - n_5 g = 0, \quad (37)$$

$$g s_{\xi\xi} + 2 g_{\xi} s_{\xi} + n_2 n_3 g_{\xi} + n_6 g^3 + n_7 g = 0, \quad (38)$$

where

$$n_3 = \frac{2}{P n_1^2}, \quad n_4 = n_3 Q_r, \quad n_5 = w n_3, \quad n_6 = n_3 Q_i, \quad n_7 = n_3 \frac{R}{2}.$$

In order to find travelling wave solutions, we shall use the ansatz (solutions) suggested by Porubov and Velarde⁴³

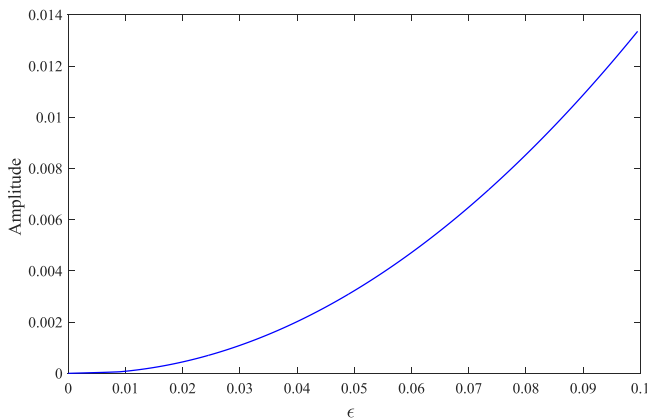


FIG. 6. Plot of the ionic wave amplitude as a function of the amplitude of the perturbation. The parametric values are the same as in Fig. 4.

$$R = g^2, \quad (39)$$

$$A = g^2 s_{\xi}. \quad (40)$$

Inserting these in Eqs. (37) and (38) yield

$$\frac{1}{2} R R_{\xi\xi} - \frac{1}{4} R_{\xi}^2 - A^2 - n_2 n_3 R A + n_4 R^3 - n_5 R^2 = 0, \quad (41)$$

$$A_{\xi} + \frac{1}{2} n_2 n_3 R_{\xi} + n_6 R^2 + n_7 R = 0. \quad (42)$$

The harmonic balance method is carried out by choosing a solution that utilizes terms which enhance a check at different orders of neglected harmonics. Thus, one needs to have an insight into the possible solution of the system in order to apply this method. The idea here is to express solutions in a periodic form at different harmonics. We therefore suppose that

$$R = r_0 + s_1 \sin l + r_2 \cos 2l, \quad (43)$$

$$A = c_0 + a_1 \sin l + c_2 \cos 2l + a_3 \sin 3l. \quad (44)$$

In terms of l , we can show that

$$R_{\xi} = f \frac{dR}{dl}, \quad A_{\xi} = f \frac{dA}{dl}, \quad R_{\xi\xi} = f^2 \frac{d^2 R}{dl^2} + f \frac{df}{dl} \frac{dR}{dl}. \quad (45)$$

Revisiting our nonlinear transform Eq. (34) and Eq. (35), we obtain $d\xi = \frac{dl}{f(l)} \Rightarrow \xi = \int \frac{dl}{f(l)}$, which leads us to the identities

$$\sec h \zeta = \cos l, \quad \text{and} \quad (46)$$

$$\tanh \zeta = \sin l. \quad (47)$$

We therefore obtain different equations at the different harmonics $\sin w l$ and $\cos y l$, where w is odd (1, 3, 5, etc.) and y is even (0, 2, 4, 6, etc.). The equations at the different harmonics are then solved to obtain expressions for the required coefficients in Eqs. (43) and (44).

From the above deductions, one obtains the factors of the Porubov and Velarde ansatz solution to the equation that describes our model as

$$R = 2r_2 \sec h^2 \zeta, \quad (48)$$

$$A = -\left(\frac{1}{2}\eta + \frac{2}{3}m \sin l\right)R - \frac{8}{3}mr_2 \sin l, \quad (49)$$

where $n_6 r_2 = m$ and $n_2 n_3 = \eta$.

We therefore obtain the ionic wave or voltage stimulus in our transmission line or model as

$$\begin{aligned} V_n(u, t) = & \left(2\epsilon\sqrt{2r_2} \sec h \zeta\right) \cos \left[\theta_n - \left(\frac{1}{2}\eta \zeta\right) \right. \\ & \left. - \left(\frac{2}{3}m\right) \ln(\cosh \zeta) - \left(\frac{1}{3}m\right) \cosh 2\zeta + wt \right] \\ & + 4\epsilon^2 r_2 x \sec h^2 \zeta \cos \left[2\theta_n - \eta \zeta - \left(\frac{4}{3}m\right) \ln(\cosh \zeta) \right. \\ & \left. - \left(\frac{2}{3}m\right) \cosh 2\zeta + 2wt \right] - 4\epsilon^2 r_2 y \\ & \times \sec h^2 \zeta \sin \left[2\theta_n - \eta \zeta - \left(\frac{4}{3}m\right) \ln(\cosh \zeta) \right. \\ & \left. - \left(\frac{2}{3}m\right) \cosh 2\zeta + 2wt \right], \end{aligned} \quad (50)$$

where

$$x = -\frac{2\omega^2 \eta_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)}, \quad (51)$$

$$y = -\frac{\omega \gamma_1 D_0}{D_0(\Omega^2 + 2\omega^2) + 2(\omega^4 - 2\omega^2 \Omega^2 + \Omega^4)}, \quad (52)$$

$$\zeta = n_1 u + n_2 t. \quad (53)$$

The voltage stimulus in Eq. (50) is that of the ionic wave propagated in our model of microtubules. This wave has its origin from the drifting of calcium ions from the region where they are more concentrated (region of higher potential) to the region where they are less concentrated (region of lower potential) within a microtubule protofilament between dimer molecules. An electric field is thus built up and is depicted by our transmission line or model.

Figure 7 presents this ionic wave on all the computational domains. It is observed that this analytical solution emerges in the form of a symmetric localized wave packet, travelling along the one-dimensional spatial subcellular network. Figures 7(a)–7(c) show its three-dimensional spatio-temporal evolution, its upper view, and its profiles at three different propagation times, respectively. This ionic wave is indicative of the existence and propagation along the array of biochemical systems of a symmetric bright envelope soliton. It is noteworthy that by definition, a bright soliton is a solution of a partial differential equation that is localized in space with the characteristic that it vanishes at infinity on the spatial domain.³⁹ Furthermore, in soliton theory, keeping in mind the concept of amplitude modulation of signal, it is well known that a carrier wave (wave front) can change its profile in the course of time since it can possibly be structurally unstable,^{37,39,40} while its corresponding envelope preserves its shape. In other words, several profiles of a carrier

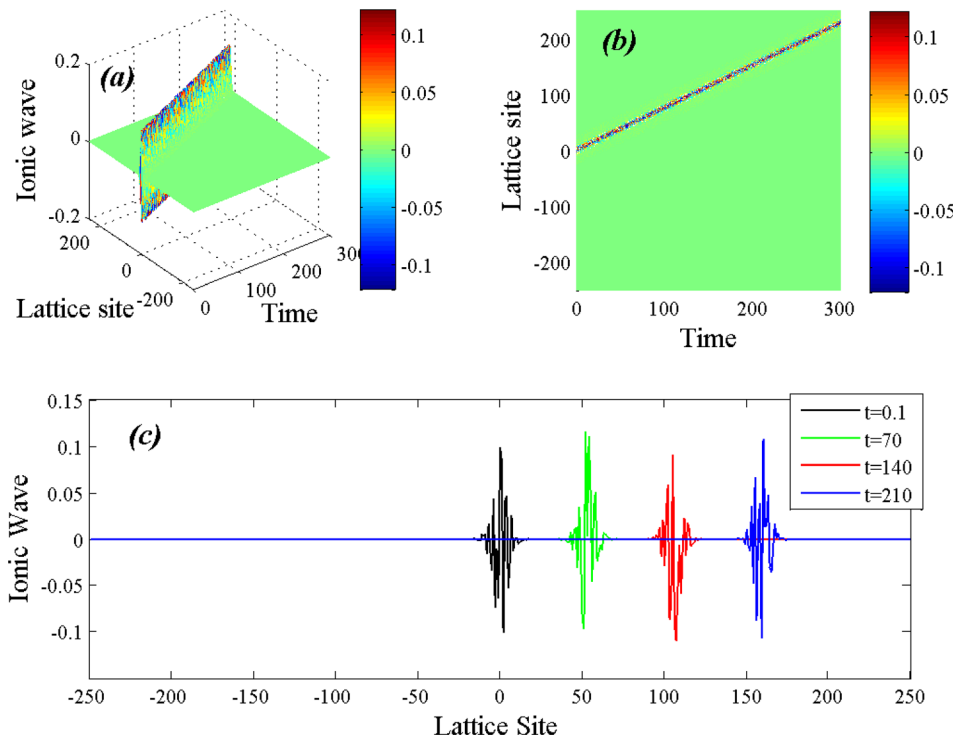


FIG. 7. Spatiotemporal evolution of the ionic wave in all the computational domains: (a) Three-dimensional view of the ionic wave, (b) upper view of the ionic wave, and (c) ionic wave profiles at different times. The signal appears in the form of a nearly symmetric wave packet. The parametric values are $D_0 = 0.04$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta = 1.0$, $\gamma_1 = 40000$, $r_2 = 1.0$, $n_1 = 1.0$, $n_2 = 0.069441$, $q = 1.5$, and $\epsilon = 0.001$.

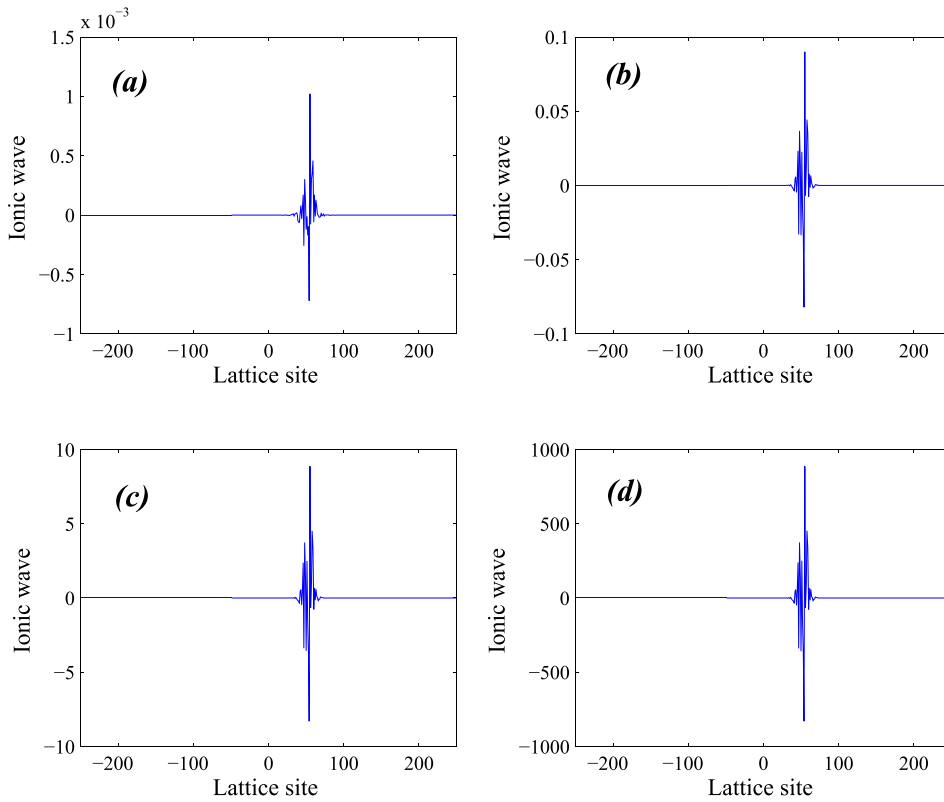


FIG. 8. Effects of small perturbations on the ionic wave. Snapshots of the wave profile taken at $t=71$. (a) $\epsilon=0.0001$, (b) $\epsilon=0.001$, (c) $\epsilon=0.01$, and (d) $\epsilon=0.1$. The parametric values are the same as in Fig. 7.

wave can support the same envelope (or amplitude). Indeed, a close look at the wave front in Fig. 7 shows that the wave packet, in the course of its propagation, retains its width and amplitude. Therefore, its corresponding envelope naturally maintains its shape as it moves in the lattice.

The impact of the perturbation on the ionic wave can be seen in Fig. 8. This figure depicts the snapshots of the internal spatial state of the microtubule at the time $t=71$, for different values of the perturbation, specifically, $\epsilon=0.0001$ [Fig. 8(a)], $\epsilon=0.001$ [Fig. 8(b)], $\epsilon=0.01$ [Fig. 8(c)], and $\epsilon=0.1$ [Fig. 8(d)]. We notice that as the perturbation increases, the amplitude of the ionic wave is tremendously enhanced. A similar trend has been reported in Sec. III in the case of the asymmetric ionic wave. A generalization of this fact is equally displayed in Fig. 9.

It therefore follows that, for very small perturbations of microtubules, impulses are exchanged between microtubules

and equally amid their inner parts in the form of nearly symmetric envelope solitons. These impulses are steadily transmitted at a constant velocity with no change in the topology of the conduction medium. Microtubules play crucial roles in higher neuronal functions including mechanisms of consciousness and memory; and it is well known that some reasonable degree of disturbances may usually occur during the events of loss of consciousness and memory in living beings. In such scenarios, an increase in the amplitude of the wave is noticed. This is as a result of the energy released from the bursting of bubbles that are formed due to the agitations stirred up in the medium by the perturbations.

Unlike the results obtained with the asymmetric solitary wave in Subsection IV A above, Fig. 7(c) shows that this symmetric ionic wave propagates in the form of a train of localized pulses embedding a series of harmonics. Therefore, this solitary wave satisfactorily meets our expectations.

V. NUMERICAL ANALYSIS OF THE VOLTAGE EQUATION: EXCITABILITY AND COLLISION

This section is solely dedicated to the direct numerical resolution of the original discrete equation of the voltage stimulus obtained in Sec. II and given by Eq. (4). The aim of this study is to check if our model of microtubule, in its full dimension, can equally display the features of excitability by supporting strongly nonlinear localized waves along its structure. For the purpose of this examination, we assume a huge partition of 100 electrical units in a one-dimensional spatial domain $[-L, L]$, with a time step of 0.1. We numerically analyze our second order ordinary differential equation based on the fifth-order Runge-Kutta scheme. We take in this case $L=50$, with periodic boundary conditions.

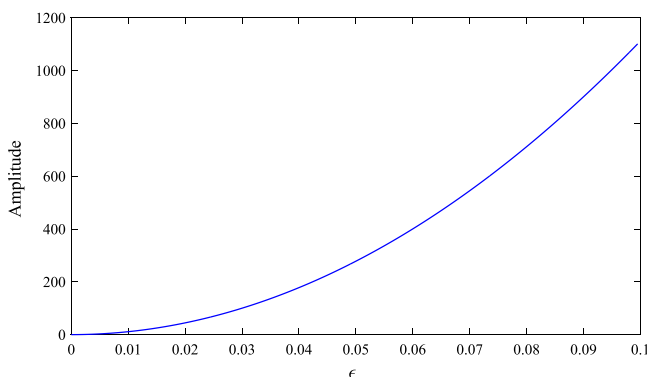


FIG. 9. Plot of the ionic wave amplitude as a function of the amplitude of the perturbation. The parametric values are the same as in Fig. 7.

This initial value problem will be resolved while assuming as Ansatz, the exact analytical solution obtained in Sec. IV as Eq. (31). An inspection of the collective dynamical behavior of our electrical network enables us to witness several distinct wave phenomena occurring in our subcellular milieu within several parametric conditions.

In the upper part of Fig. 10, it is observed that after sending the impulse in the system, the structure preserves this excitation state permanently, with no event of propagation, for some given set of the system parametric values. This suggests that stationary localized pulses are accessible for our biological system. This intimates the existence of a stationary bright envelope soliton solution of our discrete voltage equation. Figures 10(a)–10(c) show the three-dimensional spatiotemporal display, the upper view, and the profile of this ionic wave at all times, respectively.

Interestingly, a further numerical study in other parametric conditions exhibits scenarios whereby, subsequent to the insertion of the Ansatz, it breaks down into a bi-pulse (that is two pulses) propagating in opposite directions within

the lattice. This phenomenon is observed in both the lower part of Fig. 10 and in Figs. 11 and 12, for different lattice settings. Due to the periodic boundary conditions, these two pulses collide at the boundaries of the biological lattice and subsequent to this collision, several physical events may occur. First, Figs. 10(d)–10(f) disclose a situation where the two pulses merge into a non-trivial spatially homogeneous steady state for the biological ensemble. This stationary state is reminiscent of the phenomenon of oscillation quenching (observed in many physical and biological systems), alongside with the concept of synchronization as all the dynamical units states saturate at the same value of the amplitude of the ionic signal (voltage).

Second, in Fig. 11, with some additional boundary conditions, namely, the homogeneous Dirichlet boundary conditions, the two pulses after the collision, coalesce into a stable stationary dark soliton. Figures 11(a)–(11c) portray the three-dimensional spatiotemporal evolution of the bi-pulse propagation and collision, its upper view, and the profiles of the ionic wave at different times, respectively. Figures

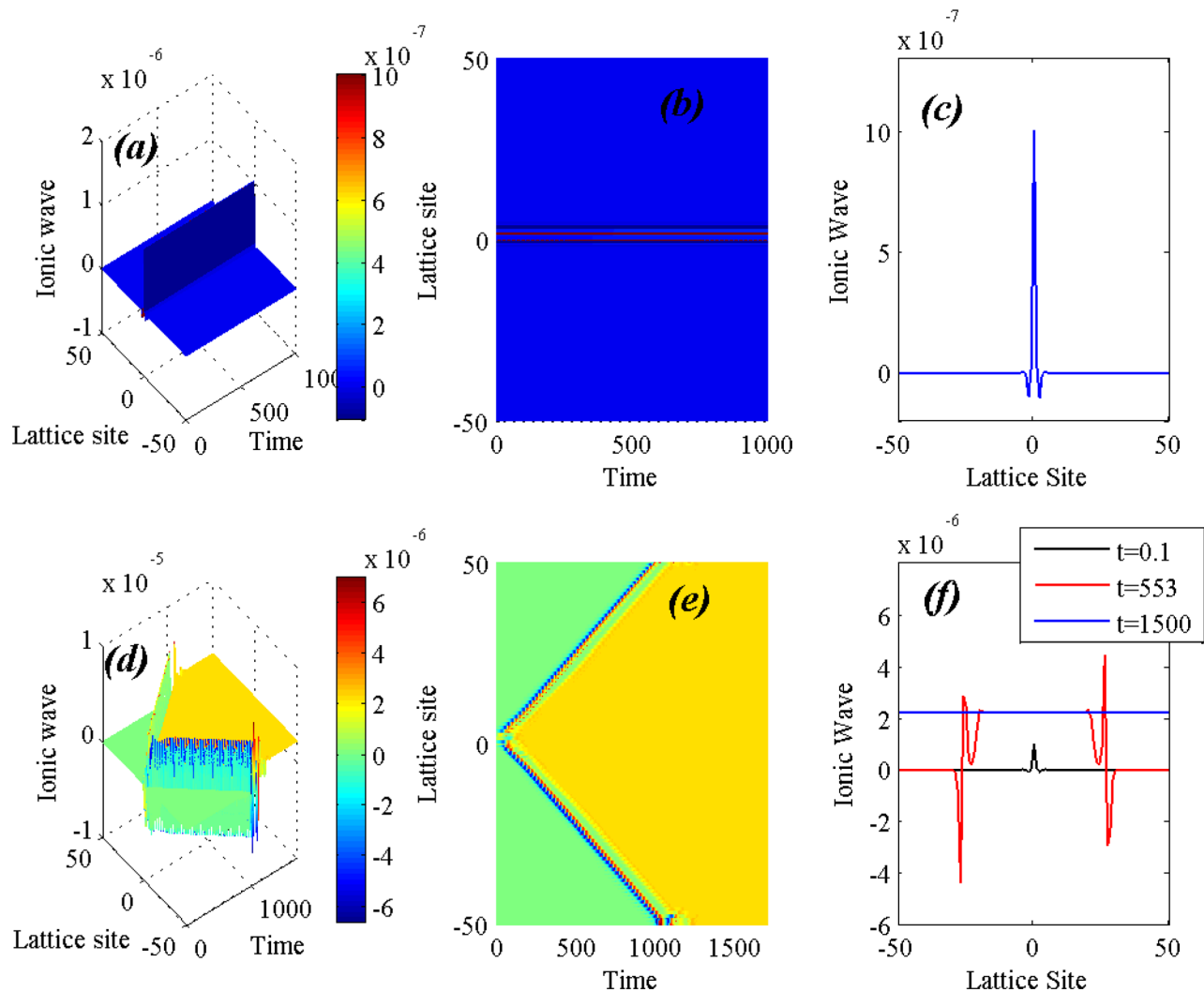


FIG. 10. Ionic waves solutions of the discrete voltage equation (4) in two different parametric conditions. The *upper figures* depict a stationary localized pulse. (a) Three-dimensional view, (b) upper view, and (c) profile at all times. The parametric values are $\gamma_0 = 65.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 0.005$, $\gamma_1 = 40000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$. The *lower figures* depict a Bi-pulse solution travelling in opposite directions, yielding after collision a non-trivial spatially homogeneous stationary pattern. (d) Three-dimensional view, (e) upper view, and (f) profiles of the wave at different times. The parametric values are $\gamma_0 = 0.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 10^9$, $\gamma_1 = 40000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$.

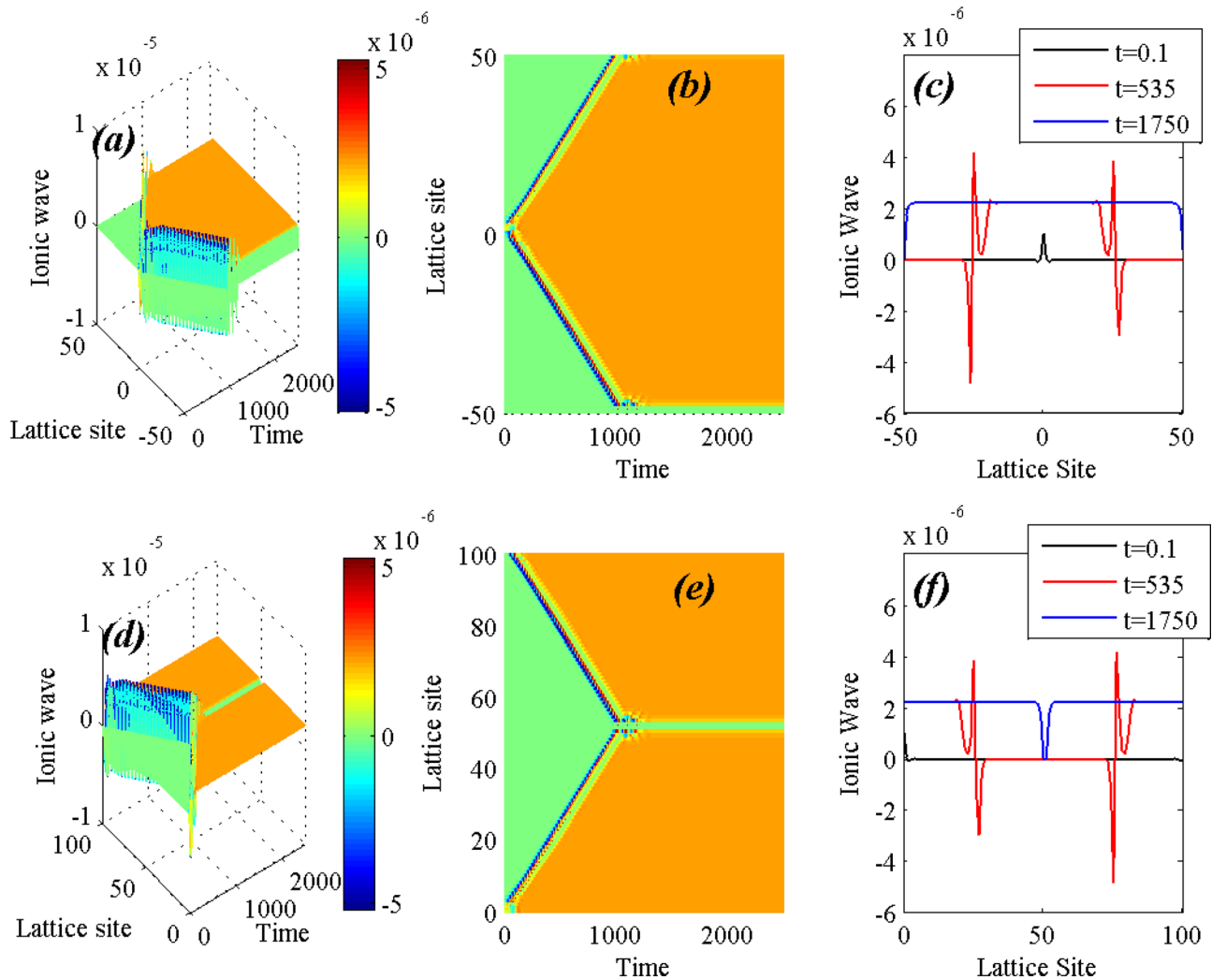


FIG. 11. Ionic waves solutions of the discrete voltage equation (4). The upper figures show a Bi-pulse solution travelling in opposite directions, yielding after collision, a stationary dark soliton. (a) Three-dimensional spatiotemporal evolution, (b) upper view, and (c) profiles at three different times. The parametric values are $\gamma_0 = 0.0$, $\Omega^2 = 0.0049$, $\eta_1 = 0.09$, $\eta_2 = 10^9$, $\gamma_1 = 40000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$. The lower figures (d), (e), and (f) depict a rotation of the cyclic lattice observed in the upper figure in order to clearly highlight the collision process of the two pulses and the emergence of the dark soliton.

11(d)–11(f) show a rotation of this large cyclic network in order to underscore the formation of the dark soliton in the lattice, after the collision process of the two propagating pulses has taken place.

Third, in other parametric conditions, Fig. 12 shows that the two pulses, as they propagate through the spatial domain, leave behind them a permanent deformation of the lattice that leans toward generating a spatially inhomogeneous stationary pattern, in the form of stripes disseminated all over the spatiotemporal domain. Figures 12(a)–12(e) depict the three-dimensional spatiotemporal evolution of the travelling ionic waves, its upper view, and the profiles of the lattice at four different times, respectively. It is important to note here that after the collision, the two pulses annihilate each other.

Therefore, the aforementioned facts stress the existence, prior to the collision process, of a robust bisoliton solution propagating along the biochemical medium.

Furthermore, in cellular automata, the state of a particular cell in the next time steps depends on the state of the cells around it (that is its neighbors) at current time. The Greenberg-Hasting cellular automaton, for instance, can be

conveniently used to model the excitability of heart tissues.⁴⁶

In general, it is well known that each cell can be in one of the following states: quiescent (or excitable) state, excited state, or refractory state. In the last case, the cell has recently been excited and is temporally not excitable. This refractory cell possesses a refractory period which is lessened at the next time step, until it reaches the end of the refractory period and becomes excitable once more. Thus, the refractory period is indeed the time interval elapsed between the time step that follows the onset of excitation, and the time step at which the cell returns to a quiescent state, when it becomes excitable again. At this stage, we numerically assess this refractory period in the context of our work. For this purpose, we select a specific cell, namely, the cell at location $n = 25$ and numerically compute its refractory period on the basis of its local dynamics as highlighted in Fig. 13.

Indeed, based on its time series observed in the same parametric conditions as in Fig. 10, we obtain a refractory period of 372.1 time units whose span is indicated by the double arrow (Magenta online). It is noteworthy that in the

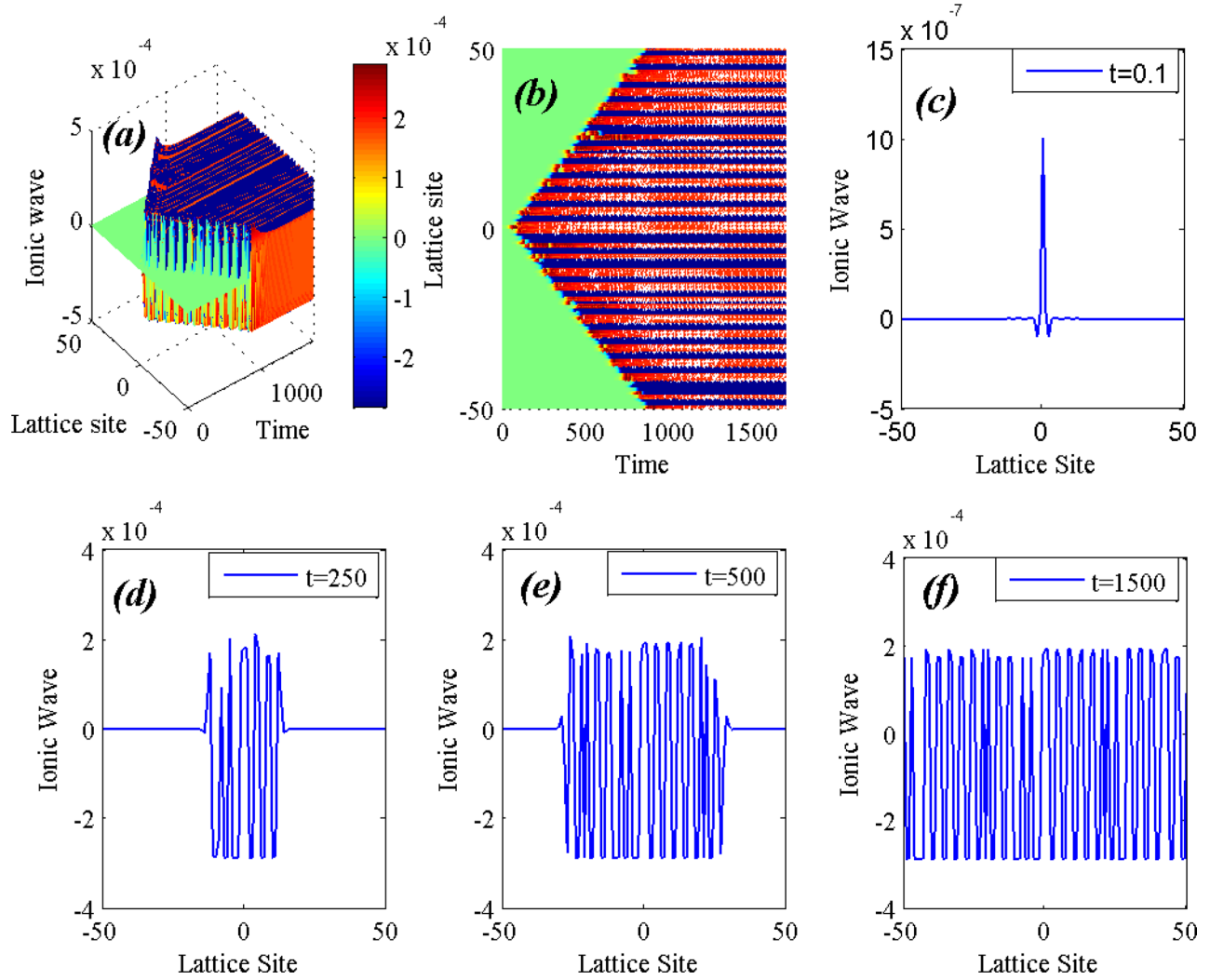


FIG. 12. Ionic waves solutions of the discrete voltage equation (4). Bi-pulse propagation, generating the emergence of a spatially inhomogeneous stationary pattern along the lattice. (a) Three dimensional spatiotemporal evolution of the bi-pulse; (b) upper view of the propagating phenomenon; (c)–(f) snapshots of the structure of the lattice at different times. The parametric values are $\gamma_0 = 65.0$, $\Omega^2 = (2.85)^2$, $\eta_1 = 0.09$, $\eta_2 = 10^8$, $\gamma_1 = 4000$, $\gamma_2 = 5.0$, $\delta = 0.1$, $\lambda = -1.0$, and $D_0 = 0.0024$.

parametric conditions of Fig. 12, this same value of the refractory period was retrieved. This fact appears to be logical due to the fact that we assumed from the onset that our lattice is made up of identical dynamical units. Therefore,

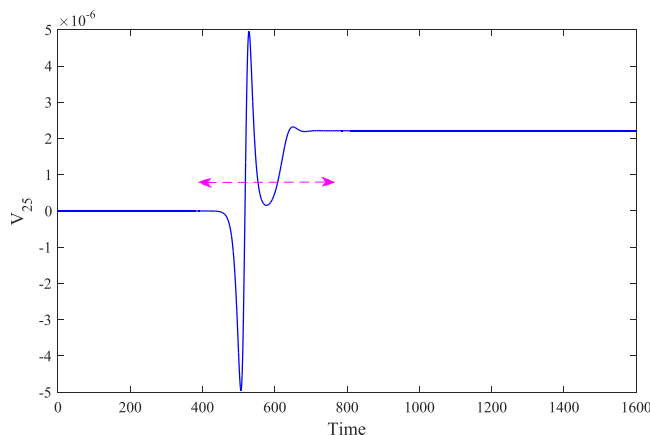


FIG. 13. Local time series of the dynamical unit at location $n = 25$, indicating the refractory period of the lattice found to be approximately 372.1 time units. The parametric conditions are the same as in Fig. 10.

based on these facts, the minimum time interval needed for any dynamical unit along our biological lattice to regain its excitability subsequent to the onset of its excitation is about 372.1 time units.

At this point of our numerical analysis, it becomes obvious that our biological system, via its electrical model of microtubule, is resiliently qualified to uphold nonlinear localized waves of excitation. This predisposition to excitability has thoroughly been proven both analytically and numerically.

VI. CONCLUSION

Excitable media are widely spread amid biological systems. They include among others, the heart, the muscular tissues, and the neural networks. In this work, we have studied the capability of the inner part of a microtubule, namely, a protofilament, to behave like an excitable structure. For this purpose, we have proposed an electrical model mimicking the behavior of a microtubule protofilament, with strongly nonlinear resistive elements. Based on this model, we could derive the discrete equation of evolution of the voltage

stimulus urging the calcium ions to flow both within the medium and in between the microtubules. We showed that this equation of motion could be reduced to a *CGLE* describing the dynamics of the amplitude of the ionic wave. An analytical and numerical treatment of this equation indicates that its plane waves solutions are continually unstable, thereby depicting the propensity of our biological milieu to favor the emergence of modulated amplitude waves. These localized modes incorporate both stationary and travelling bright solitons, bi-solitons, and dark solitons. We could equally notice the accessibility of our array of biochemical units to some stationary homogeneous and inhomogeneous spatial patterns.

Microtubule model for excitability was originally proposed for wireless communication in the brain, where microtubules at the axon initial segment may serve as the receiving antennas for converting electromagnetic waves into neuronal excitability.²¹ Some investigators hypothesized that reducing the number of microtubules with general anesthetics,²² colchicine,²³ or other antimitotic drugs²⁴ suppressed the ability of nerve cells to produce action potentials. In this case, the states of excitability analysed indicate that microtubules possess a high degree of complexity, peculiar to many nonlinear biological systems, and it plays a crucial role in the processes underlying the mechanisms of information processing and transmission in living beings at the subcellular scale.

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